

ONLINE RETAIL

EXPLORATORY DATA ANALYSIS

CodeAlpha | Task 2

Prepared by

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1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of an online retail dataset spanning December 2010 to December 2011. The dataset contains approximately 537,000 transaction records across 8 variables, covering a UK-based wholesale gift and novelty retailer operating across multiple countries.

Dataset at a Glance

536,641 total records | 8 columns | 401,604 records after cleaning | Dec 2010 – Dec 2011 | ~38 countries

After removing rows with missing CustomerIDs (135,037 records), the analysis focuses on identified customers. Key questions explored include top-selling products, revenue-generating countries, monthly sales trends, high-value customer segments, and data anomalies.

2. Dataset Overview

2.1 Structure & Data Types

The dataset contains the following fields:

Column	Data Type	Description
InvoiceNo	object	Unique invoice identifier (cancellations prefixed with 'C')
StockCode	object	Product/item code
Description	object	Product name (1,454 nulls)
Quantity	int64	Units purchased (negative = returns)
InvoiceDate	datetime64	Date and time of transaction
UnitPrice	float64	Price per unit in GBP
CustomerID	float64	Unique customer identifier (135,037 nulls)
Country	object	Customer's country of residence

2.2 Descriptive Statistics

Summary statistics for numeric columns (after CustomerID null removal):

Metric	Quantity	Unit Price	Total Price
Count	401,604	536,641	401,604
Mean	9.62	4.63	—
Min	-80,995	-11,062.06	—
Max	80,995	38,970.00	—
Std Dev	219.13	97.23	—
Median (50%)	3	2.08	—

3. Key Findings & Trends

3.1 Top-Selling Products

Analysis of quantity-based product rankings reveals the following:



Top 10 Products by Quantity Sold

- World War 2 Gliders ASSTD Designs leads with ~52,000 units — nearly 15% more than the 2nd-place product.
- Jumbo Bag Red Retrosport ranks 2nd (~45,000 units); Assorted Colour Bird Ornament is 3rd (~35,000 units).
- All top 10 products are low-cost, high-volume novelty/gift items — revenue is driven by scale, not margin.
- White Hanging Heart T-Light Holder (visible in raw data) also appears as a recurring popular item.
- Cake cases, tissue packs, and ornaments dominate — consistent with a seasonal gift/retail wholesaler.

3.2 Revenue by Country

Revenue is severely concentrated in a single market:



Geographic Revenue Distribution

- The United Kingdom generates approximately 7x more revenue than the next highest country (Netherlands).
- Top international markets in order: Netherlands, EIRE (Ireland), Germany, France, Australia.
- International markets collectively represent a small fraction of total revenue.
- Switzerland, Spain, Belgium, and Sweden show minimal but present activity.

- This single-market dependency represents a significant business concentration risk.

3.3 Monthly Sales Trend

Revenue shows a strong, predictable seasonal pattern:

Monthly Revenue Trend (2011)

- Revenue is relatively flat from January through July, ranging from approximately 0.45M to 0.65M GBP.
- A sharp upward trend begins in August, accelerating through September and October.
- November is the peak revenue month, reaching approximately 1.15M GBP — roughly 2x the January baseline.
- December shows a notable drop from the November peak, likely due to the data ending December 9, 2011.
- This classic pre-Christmas surge confirms the business is highly seasonal and gift-oriented.

3.4 Top Customers by Revenue

Customer revenue distribution reveals a classic power-law pattern:

High-Value Customer Analysis

- Customer 14646 is the single highest-value customer, generating approximately 270,000 GBP.
- Customer 18102 is a close second at approximately 255,000 GBP.
- Customer 17450 ranks 3rd at approximately 185,000 GBP.
- The top 2 customers generate approximately 2x more than customers ranked 4th through 10th.
- Heavy reliance on a small number of whale accounts creates significant revenue fragility.

4. Anomalies & Data Quality Issues

Several significant anomalies were identified that must be addressed before production reporting:

Critical: Negative Quantities & Prices

Quantity ranges from -80,995 to +80,995. UnitPrice ranges from -11,062.06 to +38,970. These negative values represent returns, cancellations, and credit adjustments — they have NOT been filtered from the analysis and are distorting all revenue figures.

Summary of Identified Issues

- 135,037 missing CustomerIDs (25.2% of records) — these were dropped, potentially understating revenue
- 1,454 missing product descriptions — minor issue, not critical
- **Negative quantities indicate return/cancellation transactions mixed into sales data**
- **Negative UnitPrices suggest credit notes or data entry errors requiring validation**
- **UnitPrice standard deviation of 97.23 with a max of 38,970 confirms extreme price outliers**
- Quantity standard deviation of 219 confirms extreme bulk orders skewing the distribution

Correlation Finding

The correlation matrix shows Quantity vs TotalPrice at 0.92 (expected and healthy). However, UnitPrice vs TotalPrice at -0.13 is counterintuitive — higher-priced items appear to generate less total revenue. This likely indicates bulk discounting on lower-price items driving volume, but warrants further investigation.

5. Actionable Recommendations

The following recommendations are prioritized based on business impact and data evidence:

#	Recommendation	Action	Priority
1	Clean Revenue Data	Filter out negative Quantity and UnitPrice rows before any business reporting. Create a validated sales_only dataframe for all KPI calculations.	High
2	Expand International Markets	Netherlands and Germany show meaningful traction. Launch targeted B2B outreach campaigns in these markets to reduce UK revenue concentration risk.	High
3	Build a Q4 Operations Plan	The August-November ramp is highly predictable. Pre-stock top-10 products 2-3 months early and scale logistics/staffing capacity before August each year.	High
4	Account Management for Top Customers	Assign dedicated account managers to Customer IDs 14646, 18102, and 17450. Offer loyalty pricing tiers. A single lost whale account represents a major revenue event.	Medium
5	Investigate Price-Revenue Anomaly	The negative UnitPrice vs TotalPrice correlation (-0.13) suggests margin erosion from bulk discounting. Conduct a	Medium

		margin analysis by product category to identify profitability drivers.	
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6. Risks & Assumptions

The following risks and assumptions should be noted when interpreting this analysis:

Data Coverage Risk

135,037 records with missing CustomerIDs were dropped from the analysis. If these represent guest or cash customers, total revenue and product popularity figures are understated. Recommend investigating whether these can be recovered from source systems.

Single-Year Limitation

The dataset covers only December 2010 to December 2011. The Q4 revenue spike cannot be confirmed as structural without year-over-year comparison data. Recommend replicating this analysis on 2012 data when available.

Negative Value Interpretation

Negative quantities and prices may include legitimate B2B credit notes and not solely data errors. Confirm with the business or finance team before removing these values from any reporting pipeline.

Customer ID Assumption

CustomerID is stored as float64, suggesting some IDs have decimal components or were improperly typed. Verify data integrity at the source before using CustomerIDs for CRM integration.

7. Conclusion

This Online Retail dataset reveals a UK-centric, seasonally-driven wholesale gift business with strong Q4 performance characteristics. While the data quality issues around missing CustomerIDs and negative transaction values require immediate remediation, the business patterns are clear and actionable.

The most pressing priorities are (1) establishing a clean, validated revenue dataset, (2) protecting and growing the top customer relationships, and (3) leveraging the predictable Q4 demand spike through proactive inventory and operational planning. International market diversification represents the most significant medium-term strategic opportunity.

With appropriate data cleaning and the implementation of the five recommendations outlined in Section 5, this business is well-positioned to improve both revenue predictability and geographic resilience.

Report prepared by **Uzair Khan** | April 5, 2026 | CodeAlpha Task 2 — Exploratory Data Analysis